



Tennessee Valley Authority, Post Office Box 2000, Spring City, Tennessee 37381

January 9, 2017

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Watts Bar Nuclear Plant, Units 1 and 2
Facility Operating License Nos. NPF-90 and NPF-96
NRC Docket Nos. 50-390 and 50-391

Subject: **Licensee Event Report 390/2017-001-00, Failure of Emergency Raw Cooling Water Pump Reverse Rotation Keys Represents Potential Common Cause Inoperability**

This submittal provides Licensee Event Report (LER) 390/2017-001-00. This LER provides details concerning failures of the Emergency Raw Cooling Water Pump motor reverse rotation clutch keys. This report is being submitted in accordance with 10 CFR 50.73(a)(2)(vii). A supplement addressing the cause of this failure and impact on safety function is expected to be submitted by March 3, 2017.

There are no regulatory commitments contained in this letter. Please direct any questions concerning this matter to Gordon Arent, WBN Licensing Director, at (423) 365-2004.

Respectfully,

A handwritten signature in black ink, appearing to read "Paul Simmons", is written over a horizontal line.

Paul Simmons
Site Vice President
Watts Bar Nuclear Plant

Enclosure
cc: See Page 2

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cc (Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Watts Bar Nuclear Plant



LICENSEE EVENT REPORT (LER)

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1. FACILITY NAME

Watts Bar Nuclear Plant, Unit 1

2. DOCKET NUMBER

05000390

3. PAGE

1 of 6

4. TITLE

Failure of Emergency Raw Cooling Water Pump Reverse Rotation Keys Represents Potential Common Cause Inoperability

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
11	10	2016	2017	001	00	01	09	2017	Watts Bar Nuclear Plant, Unit 2	05000391
									FACILITY NAME	DOCKET NUMBER
9. OPERATING MODE			11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)							
1			☐ 20.2201(b)			☐ 20.2203(a)(3)(i)			☐ 50.73(a)(2)(ii)(A)	
			☐ 20.2201(d)			☐ 20.2203(a)(3)(ii)			☐ 50.73(a)(2)(ii)(B)	
			☐ 20.2203(a)(1)			☐ 20.2203(a)(4)			☐ 50.73(a)(2)(iii)	
			☐ 20.2203(a)(2)(i)			☐ 50.36(c)(1)(i)(A)			☐ 50.73(a)(2)(iv)(A)	
10. POWER LEVEL 100			☐ 20.2203(a)(2)(ii)			☐ 50.36(c)(1)(ii)(A)			☐ 50.73(a)(2)(v)(A)	
			☐ 20.2203(a)(2)(iii)			☐ 50.36(c)(2)			☐ 50.73(a)(2)(v)(B)	
			☐ 20.2203(a)(2)(iv)			☐ 50.46(a)(3)(ii)			☐ 50.73(a)(2)(v)(C)	
			☐ 20.2203(a)(2)(v)			☐ 50.73(a)(2)(i)(A)			☐ 50.73(a)(2)(v)(D)	
			☐ 20.2203(a)(2)(vi)			☐ 50.73(a)(2)(i)(B)			☒ 50.73(a)(2)(vii)	
						☐ 50.73(a)(2)(i)(C)			☐ OTHER Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT

Dean Baker, Licensing Engineer

TELEPHONE NUMBER (Include Area Code)

423-452-4589

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
X	ERCW	CLU	Formsprag	Y					

14. SUPPLEMENTAL REPORT EXPECTED

☒ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☐ NO

15. EXPECTED SUBMISSION DATE

MONTH	DAY	YEAR
03	03	2017

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On November 10, 2016, Watts Bar Nuclear Plant personnel identified a failure of the non-reverse clutch key on Emergency Raw Cooling Water (ERCW) motor B-A. While performing a lubrication work order, it was discovered that the clutch key was sheared. Subsequent investigation identified that other clutch key failures had occurred in the recent past. The non-reverse clutch prevents the ERCW pump from rotating in the reverse direction after pump trip, which could cause the motor to develop a higher than normal in-rush current if the motor was subsequently started, such as following an accident.

Based on the potential common mode failure of the non-reverse clutch, immediate corrective actions were put in place to ensure that the safety function of the ERCW pumps to start following an accident would not be impaired. The cause of the failure is under investigation.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by internet e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Watts Bar Nuclear Plant, Unit 1	05000390	2017	- 001	- 00

NARRATIVE**I. PLANT OPERATING CONDITIONS BEFORE THE EVENT**

Watts Bar Nuclear Plant (WBN) Units 1 and 2 were at 100 percent rated thermal power (RTP).

II. DESCRIPTION OF EVENT**A. Event Summary**

On November 10, 2016, Watts Bar Nuclear Plant personnel identified a failure of the non-reverse clutch key on Emergency Raw Cooling Water (ERCW) system {EIS:BI} motor B-A. While performing a lubrication work order, it was discovered that the clutch key was sheared. Subsequent investigation identified that other clutch key failures had occurred in the recent past. The non-reverse clutch {EIS:CLU} prevents the ERCW pump {EIS:P} from rotating in the reverse direction after pump trip, which could cause the motor to develop a higher than normal in-rush current if the motor was subsequently started, such as following an accident.

Based on the potential common mode failure of the non-reverse clutch, immediate corrective actions were put in place to ensure that the safety function of the ERCW pumps to start following an accident would not be impaired. The cause of the failure is under investigation.

This event is being reported to the Nuclear Regulatory Commission (NRC) under 10 CFR 50.73(a)(2)(vii) as a potential common cause inoperability of independent trains of the ERCW system.

B. Inoperable Structures, Components, or Systems that Contributed to the Event

No inoperable systems beyond the issue with the ERCW non-reverse clutch contributed to this report.

C. Dates and Approximate Times of Occurrences

Date	Event
10/21/1985	Engineering Change Notice (ECN) 5940 issued to make clutch key weaker to avoid damage to the non-reverse assembly and motor shaft.
6/09/1999	C-A ERCW motor anti-rotation clutch and torsion arm had sheared bolts and key (CR 14894).
11/16/2015	H-B ERCW clutch key found sheared (CR 1104777). Key replaced and pump returned to service on 11/20/2015 (WO 116490727)
11/24/2015	A-A ERCW clutch key found sheared (CR 1108157).
12/17/2015	Work Order generated to inspect and replace clutch keys on ERCW motors (CR 1117187).
2/10/2016	A-A ERCW clutch key found sheared (CR 1135397). Key had not been replaced since initial identification on 11/24/2015.
5/20/2016	A-A ERCW clutch key replaced (WO 117442683)
11/10/2016	B-A ERCW found with a clutch key cracked and broken (CR 1231774). Key replaced on 11/22/16.

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Date	Event
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12/01/2016	Prompt Determination of Operability completed for ERCW Clutch key issue (CR 1236767).
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D. Manufacturer and Model Number of Components that Failed During the Event

The anti-rotation clutch is a Formsprag Overrunning Clutch, Model No. FSO-700L/2.500, Item No. CTF215H.

E. Other Systems or Secondary Functions Affected

No other systems or secondary functions were affected .

F. Method of discovery of each Component or System Failure or Procedural Error

Each clutch failure was found during quarterly maintenance of the anti-rotation clutch .

G. Failure Mode and Effect of Each Failed Component

The failure mode of the anti-rotation clutch is under investigation.

H. Operator Actions

When the common cause issue was recognized, a prompt operability determination identified corrective action required for operability.

I. Automatically and Manually Initiated Safety System Responses

There were no safety system responses associated with this issue. .

III. CAUSE OF THE EVENT**A. The cause of each component or system failure or personnel error, if known.**

The cause of the ERCW anti-rotation clutch failures is under investigation, and will be reported in a supplement to this licensee event report.

B. The cause(s) and circumstances for each human performance related root cause.

The cause of the ERCW anti-rotation clutch failures is under investigation, and will be reported in a supplement to this licensee event report.

IV. ANALYSIS OF THE EVENT

Each ERCW Pump Motor contains a non-reverse assembly on the top of the motor. The assembly is intended to keep the motor from rotating backwards. It is necessary to prevent backwards rotation since

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starting the motor during backwards rotation could cause the overcurrent relay to trip in response to a high inrush current. Also, pump/motor damage could possibly occur due to additional mechanical stresses.

The anti-reverse assembly consists of an anti-rotation clutch, and a torque arm assembly that is bolted on top of the clutch. The anti-rotation clutch only allows pump rotation in one direction. Attempted rotation in the reverse direction causes the clutch to lock up. In instances of attempted reverse rotation, the torque arm that is bolted to the clutch will rotate until it hits the torque arm stops on top of the motor. This, in turn, prevents rotation in the wrong direction.

The ERCW system has eight pumps (four pumps per train). Normally, during LOOP events, only four pumps (two pumps per train) may be loaded onto the four EDGs (one per EDG) unless a complete loss of one train of emergency power occurs at the same time leaving only two ERCW pumps. When both trains of ERCW are available, the four ERCW pumps are able to supply all required loads for all design basis events to support both units. When only two ERCW pumps are available with a loss of one train of emergency power, the two ERCW pumps are able to support the accident unit, the non-accident unit and the Spent Fuel Pool (SFP) for all design basis events except for the case of one unit shutdown within the previous 48 hours, in which case three ERCW pumps are required.

Immediately following a pump stop, the water contained in the vertical pump column will drain back to river elevation assisted by the 2 inch air release valve on the pump. During a LOOP event, the pumps will trip off. Thirty (30) seconds later the EDG will provide power to the selected ERCW pumps (10 seconds EDG start, 20 seconds ERCW pump sequencing). If the pumps selected to load onto the EDGs were the pumps previously in service, the water column may not have completely drained, and reverse rotation could occur if the anti-rotation clutch key is compromised. If the pump is rotating in the reverse direction when loaded onto the EDG, it may trip on overcurrent.

Testing performed by site personnel shows that with the anti-reverse clutch completely disabled, the pump will spin backwards and not stop spinning until approximately 55 seconds after pump trip. The actual time before the pump stops rotating in the reverse direction with a failed clutch is likely less than 55 seconds, but the actual time cannot be accurately determined.

A compensatory action has been implemented to ensure the required number of ERCW pumps have the ability to start following a LOOP. The non-running operable pumps are required to be selected by operations for auto-start. This ensures there is no water column in the pump when the start signal is received. Two operable, non-running pumps must be selected for auto-start per train.

V. ASSESSMENT OF SAFETY CONSEQUENCES

The ERCW system has eight pumps (four pumps per train). Normally, during LOOP events, only four pumps (two pumps per train) may be loaded onto the four EDGs (one per EDG) unless a complete loss of one train of emergency power occurs at the same time leaving only two ERCW pumps. When both trains of ERCW are available, the four ERCW pumps are able to supply all required loads for all design basis events to support both units. When only two ERCW pumps are available with a loss of one train of emergency power, the two ERCW pumps are able to support the accident unit, the non-accident unit and the Spent Fuel Pool (SFP) for all design basis events except for the case of one unit shutdown within the previous 48 hours, in which case three ERCW pumps are required.



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Based on recent WBN inspection information, the oldest being September 15, 2016, the clutch keys were found intact except for the B-A ERCW pump on November 10, 2016. Monitoring for reverse rotation at each shutdown and continued performance of quarterly inspection preventative maintenance give reasonable assurance of non-reverse assembly integrity. A more detailed assessment of safety consequences will be provided in the supplement to this LER.

- A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

The ERCW system is comprised, in part, of eight pumps aligned into two trains. Multiple concurrent failures are not likely.

- B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

Not applicable.

- C. For failure that rendered a train of a safety system inoperable, an estimate of the elapsed time from the discovery of the failure until the train was returned to service

Not applicable.

VI. CORRECTIVE ACTIONS

This event was entered into the Tennessee Valley Authority (TVA) Corrective Action Program and is being tracked under condition reports (CRs) 1231774 and 1236767.

- A. Immediate Corrective Actions

The failed clutch key of the B-A ERCW pump was replaced. A compensatory action has been implemented to ensure the required number of ERCW pumps have the ability to start following a LOOP. The non-running operable pumps are required to be selected by operations for auto-start. This ensures there is no water column in the pump when the start signal is received. Two operable, non-running pumps must be selected for auto-start per train.

- B. Corrective Actions to Prevent Recurrence or to Reduce Probability of Similar Events Occurring in the Future

The identification of actions is in progress.

VII. PREVIOUS SIMILAR EVENTS AT THE SAME SITE

A number of anti-rotation clutch key failures have occurred in the recent past as described in Section II.C of this LER. However, none of these failures were required to be reported to the NRC.



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VIII. ADDITIONAL INFORMATION

A supplement to this LER is anticipated to be submitted by March 3, 2017.

IX. COMMITMENTS

None.